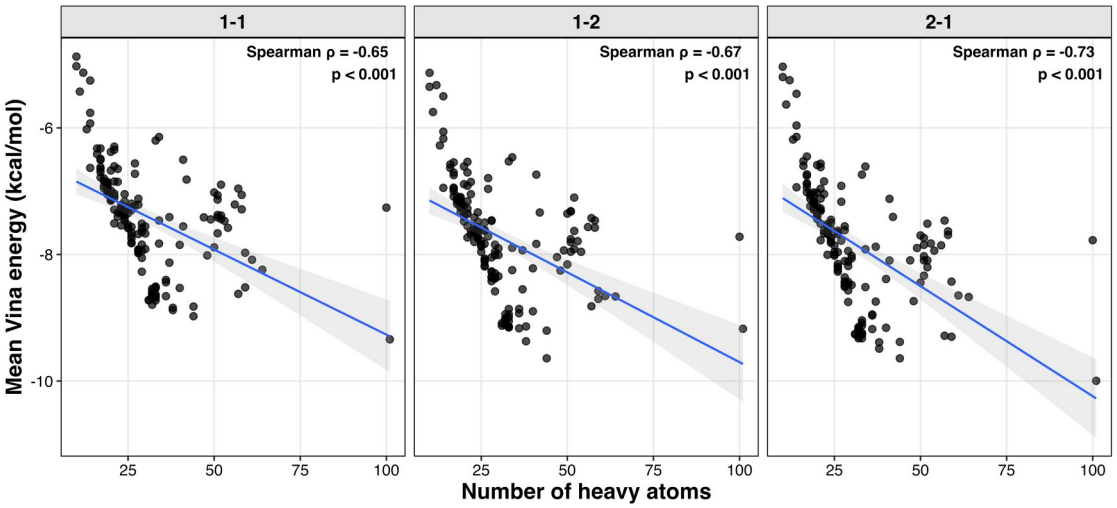


Figure S1. Relationship between the number of antibiotic heavy atoms and mean vina energy across alginate–chitosan nanoparticle formulations. Each point represents one antibiotic, and panels correspond to the 1:1, 1:2, and 2:1 Alginate–Chitosan formulations. Mean docking energy was calculated from the best-scoring pose obtained for each of the 55 nanoparticle conformation–grid combinations evaluated per antibiotic and formulation. The solid line and shaded region represent the linear regression fit and its 95% confidence interval, respectively, and are included only to visualize the overall trend. Statistical associations were evaluated using Spearman's rank correlation coefficient (rho), with the corresponding coefficient and p-value reported in each pan-el. Negative rho values indicate that antibiotics containing more heavy atoms tend to exhibit more favorable, i.e., more negative, predicted docking energies.

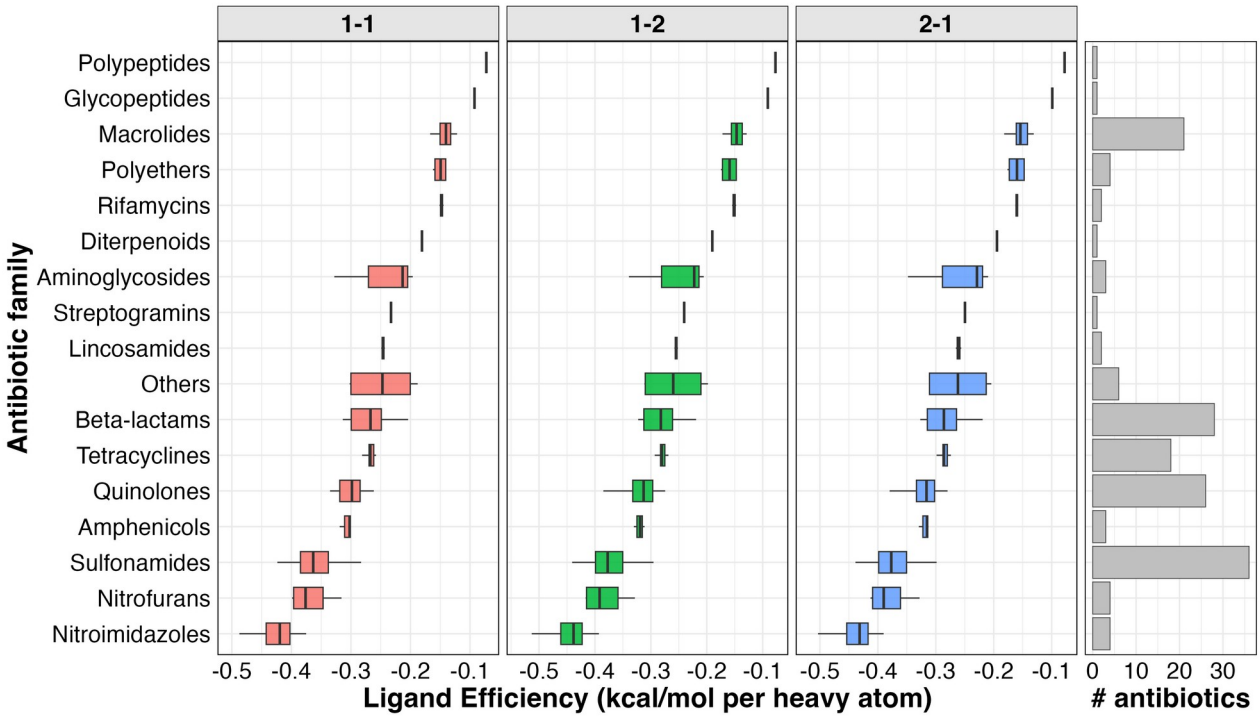


Figure S2. Distribution of mean ligand efficiency across antibiotic families for the three alginate–chitosan nanoparticle formulations (1:1, 1:2, and 2:1). Ligand efficiency was calculated by normalizing docking energy by the number of heavy atoms, providing a size-independent estimate of interaction strength. Families are ordered according to their overall median ligand efficiency, and the right marginal histogram indicates the number of antibiotics represented in each family.

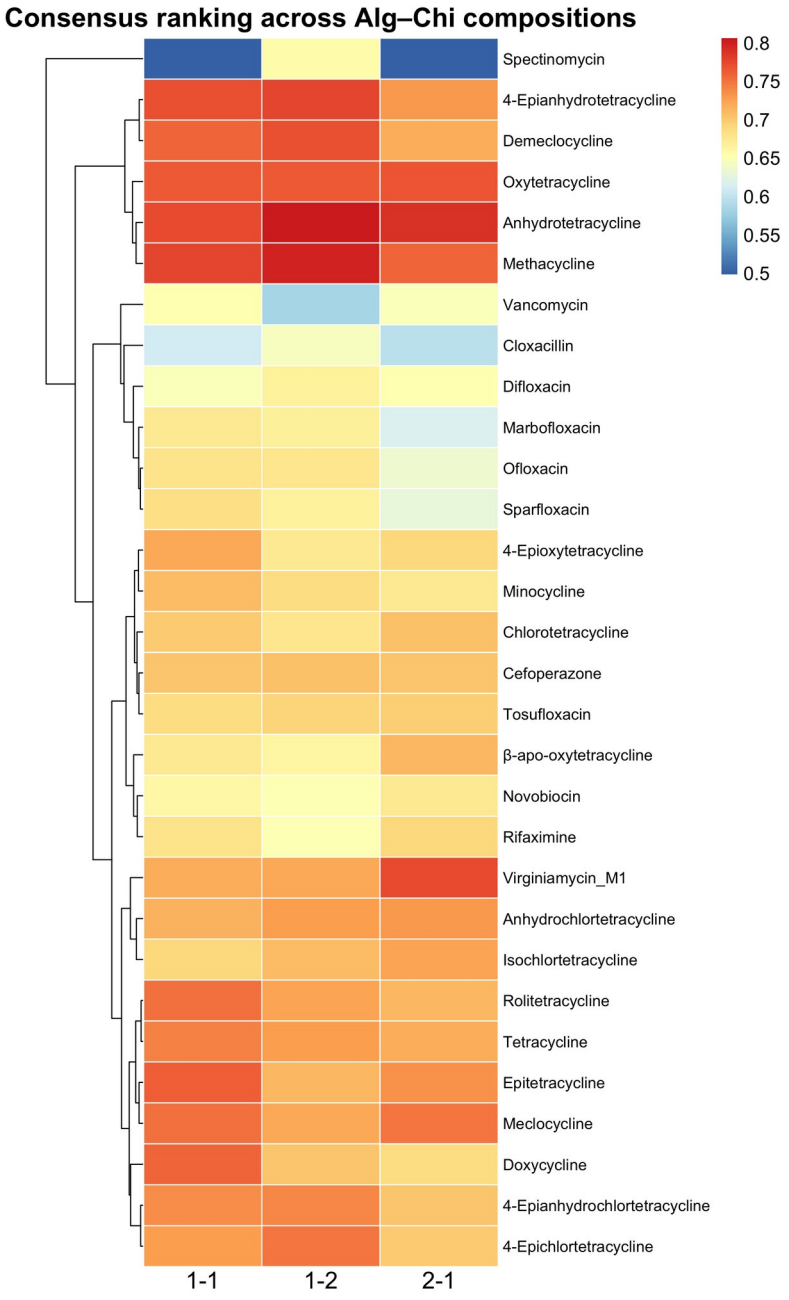


Figure S3. Consensus score heatmap for the 30 highest-ranked antibiotics across alginate–chitosan nanoparticle formulations. Rows correspond to antibiotics and columns to nanoparticle formulations. Color intensity represents the integrated consensus score obtained from docking affinity, ligand efficiency, recurrence among top-ranked docking solutions, and energetic consistency. Hierarchical clustering was performed on antibiotics to identify groups exhibiting similar prioritization patterns across formulations.

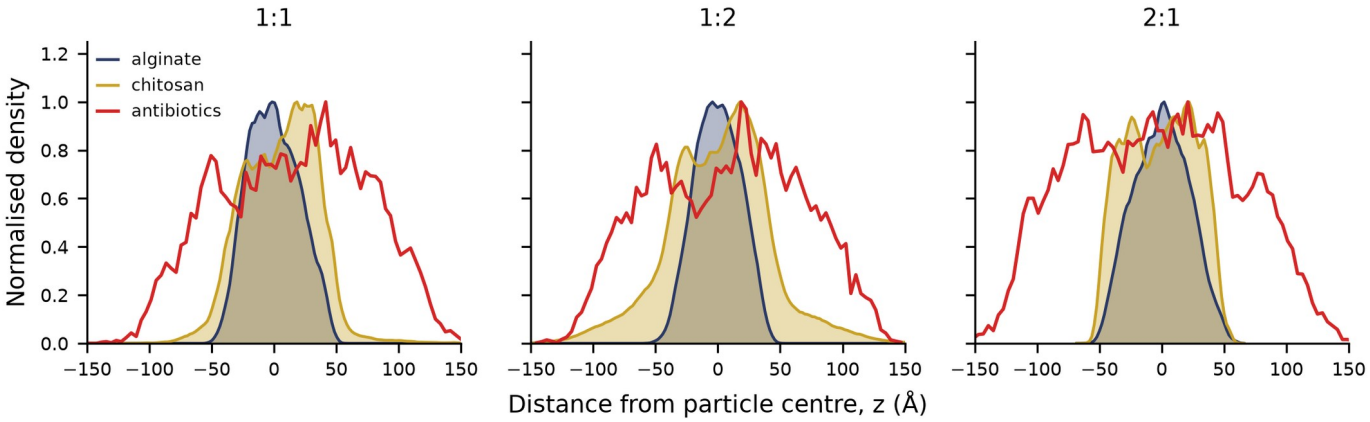


Figure S4. One-dimensional density profiles along the z axis, centred on the alginate mass centroid and normalized to unit maximum. Polymer components are shown as mass density at 2 Å resolution and antibiotics as number density at 4 Å resolution, averaged over three replicas and over all antibiotics present in each system. (a) 1:1, (b) 1:2, (c) 2:1. Chitosan extends well beyond the alginate core in the 1:2 system, consistent with the loss of chains from the assembly, whereas alginate remains compact in all three formulations.

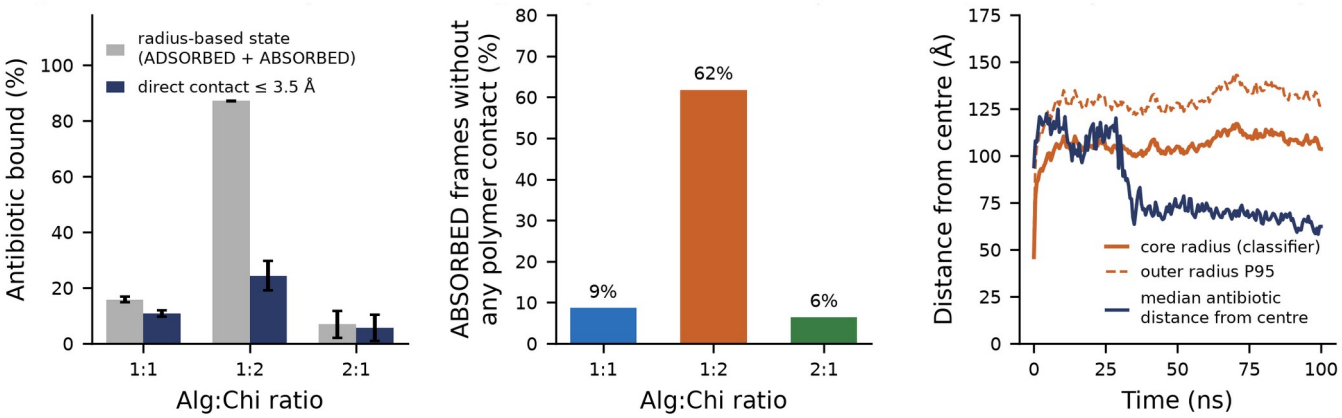


Figure S5. The radius-referenced ABSORBED classification is an artifact of particle expansion. (a) Bound fraction of antibiotic under the two definitions: the radius-referenced positional state (ADSORBED + ABSORBED) and direct interatomic contact within 3.5 Å. Bars are means across antibiotics and replicas, error bars ± 1 SD across replicas. (b) Percentage of frames classified ABSORBED in which the molecule had no atom within 3.5 Å of any polymer atom, pooled across antibiotics and replicas. (c) Representative 1:2 trajectory (replica 1, anhydrotetracycline): the classifier's core and outer radii expand past the median antibiotic distance from the particle centre within the first nanoseconds, so molecules are reclassified as internalized without approaching the polymer.